

Herald



Herald — Status

Field	Value
Revision	14
Created	2026-05-20
Last modified	2026-05-22
Status	active
	r14 captures Wave 4a (HTTP/3 + Brotli + Alt-Svc + TLS 1.3 transport substrate). v0.2.0 tag issued. Every serving flavor (pherald + cherald + sherald + iherald) now binds dual TCP/HTTP-2 + UDP/HTTP-3 listeners on the same port. Dev TLS cert auto-generated at <code>~/ .herald/dev- {cert, key} .pem</code> (ECDSA P-256, SAN localhost/127.0.0.1:::1, 365-day validity, <code>chmod 600 key</code>); prod requires <code>--tls-cert/--tls-key</code> flags or <code>HERALD_TLS_CERT_PATH/HERALD_TLS_KEY_PATH</code> env. Brotli q6 default (<code>HERALD_BROTLI_LEVEL</code> env override) compresses responses ≥ 256 B. Alt-Svc header <code>h3="<port>"</code> ; <code>ma=2592000</code> advertises HTTP/3 to TCP clients.
Status summary	<code>HERALD_DISABLE_HTTP3=1</code> escape hatch for UDP-blocked envs. New <code>commons_tls/</code> module (workspace 14 $\rightarrow$ 15). 7 new e2e invariants E49-E55 (E55 SKIP-with-reason for <code>tcpdump</code> root requirement). <code>tests/test_wave4_mutation_meta.sh</code> (4/4 PASS — M1 strip H3 $\rightarrow$ E50 FAIL; M2 strip Brotli $\rightarrow$ E53 FAIL; M3 downgrade TLS $\rightarrow$ E54 FAIL; post-flight). Critical security fix (d5bd360) restoring commons_auth/middleware.go from JWT-bypass mutation residue that the prior FIX-bg subagent's 72e81ab accidentally captured mid-flight from a concurrent mutation gate. Lesson — subagent commits must audit working-tree for mutation markers before staging. Prior r13 closed HRD-016 (Wave 3b pherald Runner live). r12 closed HRD-011 (Telegram live). r10-r11: Wave 3a substrate.
Issues	HRD-008, HRD-015, HRD-018 (in_progress), HRD-019..HRD-027, HRD-029..HRD-056, HRD-081, HRD-085..HRD-090 see <code>Issues.md</code> — 47 open items spanning §42 constitution bindings (HRD-018..027), §43 command-catalogue (HRD-029..056), HRD-081 (containers podman-compose adapter refinement), HRD-085..090 (queue-repository stub-completion surface). No HRDs migrated in r14 (substrate upgrade only). Prior cycle: HRD-016 (pherald Runner) + HRD-011 (Telegram live) + HRD-028 + HRD-098 + HRD-099 closed in r10-r13.
Issues summary	
Fixed	

Field	Value
	HRD-001..HRD-007, HRD-009, HRD-009b, HRD-010, HRD-011, HRD-012, HRD-013, HRD-014, HRD-016, HRD-017, HRD-028, HRD-080, HRD-092, HRD-093, HRD-094, HRD-095, HRD-096, HRD-097, HRD-098, HRD-099
Fixed summary	see <code>Fixed.md</code>
Continuation	Wave 4b next: TOON (Token-Oriented Object Notation) adoption as preferred Content-Type with JSON fallback. Wave 4b design lives at <code>docs/superpowers/specs/2026-05-22-http3-brotli-toon-design.md</code> §5. After Wave 4b: queued QA-bot workstream — qaherald Go binary that drives full pherald [?] Telegram round-trip automation with attached materials preserved under <code>docs/qa/<run-id>/</code> ; final close-out is the v1.0.0-dev-0.0.1 (or canonical-named) release tag. Carry-over Wave 3c items (HRD-024 iherald paging, HRD-033 destructive-guard body, HRD-018..027 flavor bindings, the rest of the §43 command catalogue) remain queued. Each milestone followed by multi-mirror push + full anti-bluff verification.

Table of contents

- [Project phase](#)
- [Specification](#)
- [Implementation](#)
- [Operations](#)
- [Risk surface](#)

Project phase

Implementation-r1, Foundation complete + Wave 2 flavor scaffolds landed. Spec V3 r8 is active, the Go module foundation compiles, all three Foundation milestones (M1/M2/M3) plus the Wave 2 flavor-scaffold workstream have landed, and the §107 end-user-usability covenant is now explicit in Herald's three root docs + gate (added 2026-05-20). The remaining cycle is wiring live channel integrations (HRD-011, HRD-016, HRD-024, HRD-028, HRD-098), validating the §26.5 quickstart end-to-end on operator hardware, and implementing the 28 §43 command bodies still stubbed.

Specification

- **Active spec:** `docs/specs/mvp/specification.V3.md` (Revision 8, ~4600 lines after r8 Wave 2 capture).
- **Archived specs:** V1 + V2 in `docs/specs/mvp/archive/` (frozen).
- **Recent sections:** §37 Tracker-doc change events; §38 Workable-item announcement contract; §39 Message presentation + Herald Canonical Template; §40 Documentation + 15 named test challenges; §41 REST API surface (Gin Gonic); §42 Constitution-binding integration; §43 Constitution-derived flavor commands (27 entries → HRD-029..HRD-056) + §43.2 catalogue-check addendum (r8); §44 Foundation implementation contract + §44.M Wave 2 milestone capture (r8); §6.3 Branding extension with per-flavor fields (r8); §18.0 Wave 2 flavor-binary catalogue (r8).
- **Spec-change rule:** §23 + HERALD_CONSTITUTION §106 + I7 gate — green throughout.

- **End-user-usability covenant:** HERALD_CONSTITUTION §107 + I8 gate (added 2026-05-20) — green; paired §1.1 mutation meta-test in `tests/test_i8_usability_meta.sh` (5/5 PASS).

Implementation

Module	Status	Notes
<code>commons</code> (L0)	✓ landed	§11.0 types + Clock + UUIDv7 + DefaultBranding; tests pass.
<code>commons_prefix</code>	✓ landed	§8.2 algorithm + 5 test functions; tests pass.
<code>commons_constitution</code> (M1)	✓ landed	14 files, ~2.9k LOC — Evaluator + Registry + 12 emit helpers + BundleHash + ModeLadder + ConstitutionStore + MemoryBus + Runner + CloudEvents v1.0 adapter; all green under <code>go test -race</code> .
<code>commons_messaging</code> (L1)	partial	<code>null://</code> adapter fully working; Telegram + Claude Code stubs in place (HRD-011/012 for live wiring). 9 SQL migrations embedded; pgx pool + RLS-enforcing <code>WithTenantContext</code> live (commit <code>c8168ec</code>); HRD-010 (commits <code>82ea82d..13cea69</code>) wired queue (<code>digital.vasic.background</code> → <code>pgxTaskRepository</code>) + Redis (<code>digital.vasic.cache</code>) + the pherald <code>migrate up/status/down/validate</code> CLI. The §107 covenant proved its worth here: the E14 RLS falsifiability test caught (and the implementer fixed) a real production RLS-bypass bug where the bootstrap PG superuser silently bypassed <code>FORCE ROW LEVEL SECURITY</code> . HRD-085..090 opened to track the remaining 16 <code>pgxTaskRepository</code> methods returning <code>ErrUnsupported</code> .
<code>commons_storage</code> (L1)	✓ landed (M2 + HRD-010)	
<code>commons_infra</code>	✓ landed	QuickstartBoot + <code>WithTenantContext</code> + on-demand container orchestration via <code>containers/</code> .
<code>commons/cli</code> (L0)	✓ landed (Wave 2 / HRD-092)	Shared CLI scaffold: <code>NewRootCmd</code> + <code>VersionCmd</code> + <code>ServeCmd</code> (with Middleware hook) + <code>StubCmd</code> + <code>healthz/readyz/metrics</code> handlers. Consumed by pherald + 6 new flavor binaries. Catalogue-check no-match → vendored Herald-internal per §11.4.74.
pherald (CLI)	✓ partial (M3 + Wave 2 refactor)	Cobra root + <code>version</code> + <code>serve</code> (live HTTP on :24791, /v1/healthz, /v1/readyz, /v1/events, /metrics) now consumes <code>cli.ServeCmd</code> with <code>RequestIDMiddleware</code> hook (commits <code>1f81d69, 31562cf</code>). <code>send/doctor/migrate/subscriber/deadletter</code> stubbed with HRD pointers.
sherald (CLI)	✓ live (Wave 3a / HRD-098)	System Herald binary serving :24793 via <code>cli.ServeCmd</code> + <code>commons_auth.GinMiddleware</code> (JWT gate). <code>/v1/safety_state</code> now LIVE — backed by <code>sherald/internal/safety</code> (process-local Aggregator + 15s mem-sampler goroutine, lifecycle bound to <code>SIGTERM</code> -cancelled

Module	Status	Notes
cherald (CLI)	 live (Wave 3a / HRD-028)	context). 5 §43 stubs (HRD-033/034/040/046/056) still return honest non-zero + HRD pointer. Constitution Herald binary serving :24792 via cli.ServeCmd + commons_auth.GinMiddleware (JWT gate). /v1/compliance now LIVE — backed by cherald/internal/compliance (paginated + filter-aware ConstitutionStore reads; tenant scope extracted from JWT claims). 11 §43 stubs still return honest non-zero + HRD pointer.
commons_auth	 landed (Wave 3a / HRD-099, renumbered from HRD-093)	JWT verifier — HMAC (HS256) + JWKS (RS256/ES256) hybrid; Redis-cached keys with in-memory fallback; Gin middleware + claims helpers. Catalogue-Check: no-match → vendored Herald-internal (docs/catalogue-checks/HRD-099-commons-auth.md).
bherald (CLI)	 scaffolded (Wave 2 / HRD-095)	Build Herald CLI-only binary. 3 §43 stubs (HRD-035/041/045).
rherald (CLI)	 scaffolded (Wave 2 / HRD-096)	Release Herald CLI-only binary. 3 §43 stubs (HRD-031/032/045).
iherald (CLI)	 scaffolded (Wave 2 / HRD-097)	Incident Herald binary serving :24794 via cli.ServeCmd. /v1/webhooks/page returns honest 501 + HRD-024.
scherald (CLI)	 scaffolded (Wave 2 / HRD-097)	Scheduled-audit Herald CLI-only binary. 1 §43 stub (HRD-047).

Operations

- **Repo hygiene:** main only; four-mirror fan-out proven across spec + Foundation M1/M2/M3 + anti-bluff infra + §107 covenant landing.
- **Constitution inheritance gate: 15 PASS / 0 FAIL** (was 12; I8a/b/c added 2026-05-20 for §107 covenant).
- **Anti-bluff audit: 16 PASS / 0 FAIL / 1 SKIP** (was 14; HRD-011 telebot.v3 third-party SKIP-with-reason per §11.4.74; I8 paired meta-test included).
- **CodeGraph validate:** 7 PASS / 0 FAIL / 2 SKIP (HRD-091 documented submodule-traversal gap).
- **E2E bluff-hunt against real services: 41 PASS / 0 FAIL / 5 SKIP** (was 33; Wave 3a added 8 new PASSES: E35/E36 auth gate × 2 flavors + E43/E44 cherald compliance live + E46/E47 sherard safety_state live; updated E28/E29 from "501 stub" → "401 JWT-gated" to reflect Wave 3a route swap). SKIPS are E17/E18/E34 (Telegram live, claude live, full inbound — requires operator credentials) + E45 (cross-binary cherald-reads-pherald-writes — Wave 3b once Runner lands) + E48 (sherard destructive-op trigger — HRD-033 body not implemented). Real Gin server, live /v1/* endpoints, real Postgres + Redis containers via containers/ submodule, M2 RLS integration tests + HRD-010 storage tests against the live stack, SIGTERM-graceful-shutdowns. Paired §1.1 mutation gates: tests/test_wave2_mutation_meta.sh (4/4 PASS — Wave 2 invariants) + tests/test_wave3_mutation_meta.sh (3/3 PASS — Wave 3a invariants M1+M5+post-flight, M6 SKIP). This is the canonical §107 evidence per Herald §107.

- **Submodules:** 12 vendored — 11 Helix-stack capability modules under `submodules/` (`auth`, `background`, `cache`, `config`, `database`, `eventbus`, `middleware`, `observability`, `recovery`, plus `Models + Concurrency` added during HRD-010) referenced via `replace` directives in consuming `go.mod` files, NOT via `go.work`; plus `containers/` (runtime auto-detection + on-demand container boot, consumed directly by Foundation tests + `pherald doctor`). All 12 carry the §11.4 anti-bluff anchor.
- **Build / test toolchain:** Go 1.25+ (verified on 1.26 per CLAUDE.md), Pandoc 3.9, WeasyPrint 66, Podman 5.8 — verified locally.

Risk surface

- **Submodule-catalogue discovery (operator mandate 2026-05-20).** Every new HRD MUST carry a Catalogue-Check: `reuse|extend|no-match <org/repo>@<sha>` line per Universal §11.4.74. New HRDs (HRD-029..056 §43 catalogue) currently default to TBD pending operator review.
- **Versioning discipline (operator mandate 2026-05-20).** Minor changes bump secondary version, major bump primary. Currently at V3 r7→r8 within the V3 series; a full V4 rewrite is reserved for a primary-version jump.
- **Live channel integrations untested at runtime.** HRD-011 (Telegram telebot live) requires operator-supplied credentials per `docs/CONTINUATION.md`. HRD-012 (Claude Code `claude --resume` live) is closed with live PASS evidence captured. The §107 covenant blocks claims of "done" on these until end-user evidence is captured.
- **HRD-008 operator-side quickstart not yet validated end-to-end on a fresh laptop.** The compose stack works in dev; operator hardware validation is the next concrete §107 evidence point.
- **No CI yet.** No GitHub Actions / GitLab CI / etc. Operator runs `scripts/e2e_bluff_hunt.sh` manually for now — that script IS the local CI surrogate.

Closed risks

- **Production RLS-bypass bug (closed 2026-05-20 during HRD-010 step 3, commit 5c1c022).** Until that commit, `commons_infra.WithTenantContext` set the tenant GUC but `ROW LEVEL SECURITY` was not FORCED on the targeted tables — and the bootstrap PG user used in tests was a superuser, which bypasses RLS regardless. The E14 falsifiability test (real Postgres, cross-tenant `SELECT` must return EXACTLY 0 rows from the other tenant) caught the bluff before any live deployment. Fix: switched the application connection to a non-superuser role + ran `ALTER TABLE ... FORCE ROW LEVEL SECURITY` in migration 000003. The §107 covenant directly prevented this from shipping; the lesson is "EXACT-N row count, never >=N" for any isolation test.

Recent activity

Wave 3a — substrate + 2 live REST routes (closed 2026-05-21)

8-step workstream commits `dbbea95` . . this shipped:

- `commons_auth/ JWT verifier scaffold` (HRD-099, renumbered from HRD-093; commit `dbbea95`) — new top-level Go module: HMAC (HS256) + JWKS (RS256/ES256) hybrid verifier with Redis-cached keys + in-memory fallback; Gin middleware (`GinMiddleware`) that gates every request behind the verifier and sets `ContextKeyClaims` on success; `claims.go` helpers (`TenantFromClaims`,

SubjectFromClaims); verifier_test.go exercises both modes incl. expired-token + wrong-signature paths. Catalogue-Check: no-match → vendor Herald-internal (docs/catalogue-checks/HRD-099-commons-auth.md).

- **events_processed migration** (commit 3f8daed) — commons_storage/migrations/000012_events_processed.{up,down}.sql; RLS-enforced 30-day TTL inbound idempotency archive per V3 §32.2.
- **ConstitutionStore.ListQuery extension** (commit 6276437) — Since/Until/Offset fields added; both Memory + Postgres backends honour them.
- **cherald/internal/compliance/Handler + tests** (commit a2ef831) — paginated + filter-aware /v1/compliance handler; tenant scope from JWT claims; defaults page=1, page_size=50 (max 200).
- **cherald wiring + route swap** (HRD-028 close-out, commit bfca740) — cherald/cmd/cherald/main.go builds the Verifier from env, instantiates the ConstitutionStore (Postgres if HERALD_PG_DSN else Memory), and wires commons_auth.GinMiddleware + compliance.Handler via cli.ServeOpts.Middleware. The 501-stub for /v1/compliance is gone.
- **sherald/internal/safety/Aggregator + Handler + sampler** (commit 1016b96) — process-local Aggregator tracks lastMemPercent + lastDestructiveOp + uptime; StartMemSampler goroutine ticks every 15s; Handler returns the snapshot as JSON; tests cover the Aggregator surface.
- **sherald wiring + sampler start** (HRD-098 close-out, commit bfa82bc) — sherald/cmd/sherald/main.go instantiates the Aggregator + starts the sampler under a SIGTERM-bound context; wires the Verifier + Handler via cli.ServeOpts.Middleware. The 501-stub for /v1/safety_state is gone.
- **e2e + mutation gate + docs sync (commit 0cc6fad)** — 8 new e2e invariants (E35/E36/E43/E44/E46/E47 PASS; E45/E48 SKIP-with-reason); E28/E29 updated from "501 stub" to "401 JWT-gated"; e2e header set process-wide HERALD_AUTH_MODE + CODEGRAPH_ALLOW_UNSAFE_NODE; new paired §1.1 mutation gate tests/test_wave3_mutation_meta.sh (3/3 PASS); atomic Issues→Fixed for HRD-028, HRD-098, HRD-099 (commons_auth, renumbered from HRD-093 in this r11 cleanup); Status.md r9 → r10 → r11; Issues.md r8 → r9 → r10; Fixed.md r7 → r8 → r9.

Anti-bluff battery sign-off (this commit): inheritance gate 15/15 PASS; meta-tests (I6, I8, Wave2, Wave3) all green; audit_antibluff.sh 16 PASS / 0 FAIL / 1 SKIP (commons_auth not added to the audit's submodules/ walk — workspace-internal module by design); codegraph_validate.sh 7 PASS / 0 FAIL / 2 SKIP (with CODEGRAPH_ALLOW_UNSAFE_NODE=1 for Node 25 environment); e2e_bluff_hunt.sh 41 PASS / 0 FAIL / 5 SKIP.

Wave 2 — flavor-scaffold sweep (closed 2026-05-21)

15-step workstream commits 7e0a614..eef606b (with parallel logo branding 24b96f2) shipped:

- **Shared scaffold** — new commons/cli/ package: NewRootCmd (steps 2), VersionCmd (step 2), routes.go + healthz/readyz/metrics handlers (step 3), ServeCmd with Middleware-hook + graceful shutdown (step 4), StubCmd honest 501-style stubs (step 1). Vendored Herald-internal per §11.4.74 catalogue-check no-match → see docs/catalogue-checks/HRD-092-commons-cli.md.
- **pherald refactor** — main + stubs migrated to consume commons/cli/ (step 5); pherald serve consumes cli.ServeCmd with RequestIDMiddleware hook (step 6, commit 31562cf).
- **6 new flavor binaries** — sherald serving :24793 (step 7, HRD-093), cherald serving :24792 (step 8, HRD-094), iherald serving :24794 (step 9, HRD-097), bherald CLI-only

(step 10, HRD-095), rherald CLI-only (step 11, HRD-096), scherald CLI-only (step 12, HRD-097). Each ships honest 501-style §43 command stubs with HRD pointers.

- **Anti-bluff additions** — `scripts/e2e_bluff_hunt.sh` grew from 18 → 33 invariants covering all 6 new binaries (version --json shape, healthz binding, /v1/safety_state + /v1/compliance + /v1/webhooks/page 501 honesty, sherald SIGTERM graceful-shutdown, §43 stub exit-code, pherald regression sentinel). New paired mutation gate `tests/test_wave2_mutation_meta.sh` (4/4 PASS) proves the new invariants catch their claimed regressions.
- **Spec capture** — V3 r7 → r8 added §6.3 Branding extension (per-flavor fields), §18.0 Wave 2 catalogue, §41 REST route additions, §44.M Wave 2 milestone, §43.2 catalogue-check addendum (step 14, commit `ee f606b`).
- **Atomic Issues→Fixed** — HRD-092..097 closed in r7 of Fixed.md; HRD-098 opened in r8 of Issues.md as Wave 3+ deferred (sherald /v1/safety_state live impl).

Anti-bluff battery sign-off (this commit): inheritance gate 15/15 PASS; meta-tests (I6, I8, Wave2) all green; `audit_antibluff.sh` 16 PASS / 0 FAIL / 1 SKIP; `codegraph_validate.sh` 7 PASS / 0 FAIL / 2 SKIP; `e2e_bluff_hunt.sh` 33 PASS / 0 FAIL / 3 SKIP.